APPLICATION BLUEPRINT AND OPERATING GUIDE

# iSurvix Field Application Guide

Purpose • Delivered features • People and permissions • Review controls

This guide defines how iSurvix Field should operate and records what has been delivered in the local application. It gives the project owner, administrators, Engineers, Team Leads and Clients one reference for setup, day-to-day work, approvals and the next release.

The working baseline supports invitation-based project access, custom checklist selection, Engineer submission, Team Lead review, optional Client approval and local AI photo drafts. Human decisions control acceptance and closure. Real-site photo accuracy and production operation still require validation.

| Document control | Baseline |
|---|---|
| Version and date | 1.0 • 17 October 2026 |
| Application location | F:\iSurvix\iSurvix-Field |
| Local entry point | http://127.0.0.1:8000/local/ |
| Audience | Project owner, administrators, delivery teams and Clients |
| Evidence basis | Current source, specialist reviews, automated tests and local synthetic pilot |
| Handling | Application guide; account identifiers included; passwords excluded |

## Reading map

| Pages | Subject |
|---|---|
| 2–4 | Delivered scope, role responsibilities and permission matrix |
| 5–7 | User directory, administration and review workflow |
| 8–10 | Custom checklists, photo evidence and AI controls |
| 11–12 | Reports, audit, security and local operations |
| 13–14 | Validation, release plan and source references |

**Brand treatment follows the app: forest green #286D3D, dark ink #20382F, white surfaces and soft green #EAF2E9, with the existing iSurvix logo.**

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# Purpose and delivery status

The app should carry each inspection activity from assignment to evidence-backed completion while preserving its checklist version, approval history and project boundaries.

## Operating model

Super Admin creates projects and assigns primary Project Admins. Each Project Admin builds the project team through invitations. A Team Lead assigns an activity to an Engineer. The Engineer records answers and evidence, reviews AI suggestions and submits. The Team Lead reviews the work and AI evidence. A configured Client makes the final approval before closure; otherwise Team Lead approval closes the activity.

| Capability | Current position |
|---|---|
| Authenticated workspace | Delivered locally. The sample-data website route is retired; its source remains for reference. |
| Project governance | Delivered: projects, primary Admin assignment, scoped invitations, membership controls and audit. |
| Checklist builder | Delivered: full master, section/item selection, fresh drafts, publication, assignment and frozen survey snapshots. |
| Review workflow | Delivered: assigned Engineer → actual Team Lead → configured Client → closed; no-Client branch closes after Team Lead. |
| Photo processing | Delivered pilot: validated upload, original preservation, durable queue, local vision, draft review and JSON evidence report. |
| Local model | qwen3.5:4b vision capability and native image requests verified; synthetic end-to-end pilot passed. |
| Release readiness | Local testing baseline. Real-site accuracy, sustained bulk performance and production acceptance remain open. |
| Future capability | Measured sketches, editable SVG/DXF, external delivery and wider deployment are planned, not delivered. |

## Data and product boundaries

Existing answers, original photos, published checklist revisions and audit history must survive normal upgrades. The current application is local and loopback-only. A separate hosted redesign is not a live connection to these local records. No public hosting removal or production deployment is claimed by this guide.

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# Roles and responsibility

Five roles are currently supported. Authority comes from the authenticated account and its project scope, rather than a role selected at sign-in.

| Role | Accountability and scope |
|---|---|
| Super Admin | Owns application governance across projects. Creates projects, invites Project Admins, assigns the primary Admin, inspects global audit events and performs access maintenance. |
| Project Admin | Owns team access and checklist setup in one project. Invites Engineers, Team Leads and Clients; manages permitted membership controls, project work and scoped checklist revisions. |
| Team Lead | Owns delivery coordination and internal review in the project. Assigns activities to Engineers, reviews submissions and AI evidence, approves sections or returns work for correction. |
| Engineer | Owns execution of assigned activities. Enters checklist answers, uploads evidence, accepts or leaves AI suggestions, corrects editable sections and submits the work. |
| Client | Owns the Client approval decision. Reads final project submissions and evidence, exports deliverables and approves or rejects an open Client review. Cannot change engineering answers. |

## Stage signatures remain separate

Only the actual assigned Engineer can submit. Only an actual project Team Lead can sign the internal review. Admin and Super Admin permissions do not substitute for either signature. Only an authenticated Client approval can close an activity whose Client gate is required. AI cannot sign any stage.

## Administrative assistance

Team Leads and administrators can correct unlocked fields of non-Final surveys within their scope, including submitted or review states. This support access does not grant submission or review signatures. Engineers work on their own editable Draft, Rejected or Partially approved checklists. Full Team Lead rejection can clear section approvals only before internal Final, while Submitted, in Review or Partially approved. Final records remain locked.

## Primary Admin designation

A project has a primary Admin designation. Replacing that designation preserves existing legitimate project memberships. Super Admin must separately deactivate an old Admin when access should end. An Admin already assigned to another project cannot be silently moved by the primary assignment control.

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# Permission matrix

Project scope applies to every allowed action. Engineer scope is restricted further to assigned work. All accounts must be active, authenticated and unlocked.

| Action | Engineer | Team Lead | Client | Admin | Super |
|---|---|---|---|---|---|
| Read working surveys | Assigned | Project | Final only | Project | All |
| Correct unlocked non-Final fields | Own editable | Project | No | Project | All |
| Submit Engineer work | Assigned only | No | No | No | No |
| Sign internal review | No | Project | No | No | No |
| Sign Client approval | No | No | Project open Final | No | No |
| Upload and process photos | Editable own | Editable project | No | Editable project | Editable all |
| Accept AI answer drafts | Editable own | Editable project | No | Editable project | Editable all |
| Read final evidence and export | Assigned | Project | Project Final | Project | All |
| Assign work to Engineers | No | Project | No | Project | All |
| Create checklist versions | No | No | No | Scoped | All |
| Invite Engineer Lead Client | No | No | No | Project | All |
| Invite Project Admin | No | No | No | No | Yes |
| Create project assign primary Admin | No | No | No | No | Yes |
| View governance audit | No | No | No | Project | Global |
| Run access maintenance | No | No | No | No | Yes |

## How to interpret the matrix

“Editable” means the checklist and field are writable under the current status and section approval locks. A broad read permission never unlocks a Final checklist. Clients have no photo upload, inference queue, retry or draft acceptance permission. Read-only text AI context may be allowed for their own Final records; it cannot mutate a checklist.

Project Admin membership controls include activation, deactivation, unlock and permitted Engineer/Team Lead role changes. Client role conversion is blocked. Super Admin status requires an actual Django superuser account; a profile label alone cannot grant it. Invitations cannot create another Super Admin.

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# Current user directory

The current database contains 16 active accounts: ten Engineers, three Clients, two Team Leads and the existing Super Admin. Training accounts belong to iSurvix Demo Training. This inventory was checked on 17 October 2026.

| Username | Role | Project or scope |
|---|---|---|
| super_assim | Super Admin | All projects |
| demo_eng01 | Engineer | iSurvix Demo Training |
| demo_eng02 | Engineer | iSurvix Demo Training |
| demo_eng03 | Engineer | iSurvix Demo Training |
| demo_eng04 | Engineer | iSurvix Demo Training |
| demo_eng05 | Engineer | iSurvix Demo Training |
| demo_eng06 | Engineer | iSurvix Demo Training |
| demo_eng07 | Engineer | iSurvix Demo Training |
| demo_eng08 | Engineer | iSurvix Demo Training |
| demo_eng09 | Engineer | iSurvix Demo Training |
| demo_eng10 | Engineer | iSurvix Demo Training |
| demo_client01 | Client | iSurvix Demo Training |
| demo_client02 | Client | iSurvix Demo Training |
| demo_client03 | Client | iSurvix Demo Training |
| demo_lead01 | Team Lead | iSurvix Demo Training |
| demo_lead02 | Team Lead | iSurvix Demo Training |

## Account handling

These fifteen demo accounts were created through invitation acceptance and checked through normal sign-in. They are training identities, not evidence that real people have reviewed field work. Their contact addresses use the reserved example.test domain. The listed Super Admin password was reset separately at the owner’s request.

Temporary credentials remain in private local credential files and are intentionally excluded from this guide. Use the previously supplied credential table for testing. Do not circulate a manual containing working passwords. Replace temporary passwords before any broader use and deactivate training accounts when they are no longer needed.

## Project Admin onboarding

No new named Project Admin is invented in this directory. Super Admin can invite a real Project Admin to the selected project, then assign that person as primary Admin. User and project membership can change after this dated baseline; People & invitations is the current operational record.

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# Project and team administration

Provision the project and its accountable administrator before assigning inspection activities. Each invitation carries its role and project; the invited person does not choose a role at sign-in.

| Step | Responsible person | Required outcome |
|---|---|---|
| 1 Create project | Super Admin | A named project exists with the intended scope. |
| 2 Invite or select Admin | Super Admin | An eligible Project Admin accepts the project invitation or is selected from eligible existing Admins. |
| 3 Assign primary Admin | Super Admin | Projects & admins shows the primary administrator. Empty primary assignment is filled on valid Admin invitation acceptance. |
| 4 Invite team | Project Admin | Engineers, Team Leads and Clients are invited into this project only. |
| 5 Prepare checklist | Project Admin | A published, appropriate checklist revision is assigned to the project. |
| 6 Assign activity | Team Lead or Admin | An Engineer in the project receives the intended site checklist and task. |
| 7 Confirm Client route | Project Admin | Client accounts or valid invitations reflect the intended approval policy before internal completion. |

## Invitation controls

Private invitation links are shown once for manual distribution. They expire after 72 hours, are single-use and require the invited email address. Only a SHA-256 token hash is stored. Acceptance rechecks the issuer’s current role, project authority, active state and lockout. A replacement invitation supersedes an older open invitation for the same email and project.

Automatic email sending and an independent email-ownership challenge are not implemented. The local link works on this computer while loopback-only access is in effect. Do not treat a manually distributed local invitation as a deployed external onboarding service.

## Access changes

Deactivate membership when a person should stop accessing the project. Unlock only after the cause is understood. Replacing the primary Admin does not remove another Admin’s existing membership. Keep audit evidence of access changes and avoid deleting project history to resolve an account issue.

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# Activity review and closure

Internal Final status and activity Closed status mean different things. Final means all internal sections are approved and engineering answers are locked. Closed means the configured approval chain is complete.

| Stage | Decision and next state |
|---|---|
| Engineer | Completes assigned checklist and evidence, explicitly accepts suitable AI drafts, then submits. |
| Team Lead | Reviews the submission and source evidence. Full approval, or approval of the last outstanding sections, produces internal Final. |
| Client configured | Activity becomes Awaiting client approval. A signed-in Client approval closes it. |
| No Client configured | Team Lead completion closes the activity immediately. |
| Client rejection | Requires a reason or reference; the activity stays open at Client review and internal answers remain locked. |

## Client policy is frozen at internal completion

At the first complete internal approval, the app records whether a Client is configured. Any Client account assigned to the project counts, including an inactive or temporarily locked account. A valid, unexpired outstanding Client invitation with a valid issuer also counts. Removing or deactivating a Client later does not bypass a required review. Adding a Client later does not reopen a correctly closed no-Client activity.

## Return and correction paths

Partial approval leaves incomplete work open and locks approved sections. A request for changes preserves existing section locks. Full Team Lead rejection can clear section approvals and return work for correction only while Submitted, in Review or Partially approved. It cannot reset an internally Final record. Client rejection records the rejection and keeps the Client gate open with engineering answers locked. A later fresh Client approval may complete the activity.

## Decision integrity

The app rechecks role, project, current data and signed review snapshot before committing a decision. Stale decisions are rejected. Closed activities reject subsequent Client decisions. Admins may record external correspondence, but that record cannot close a required authenticated Client stage. Historical completed records remain completed; the routing migration does not invent past approvals or rewrite their data.

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# Checklist design and customization

Use the full checklist as the controlled source, then select only the sections or items needed for a project. A custom checklist becomes a new draft; it does not edit the master or existing survey snapshots.

## Master structure

The canonical FDED source contains 172 fixed Excel rows plus inventory field nodes, represented as 181 field nodes across 16 sections. These are different counts. The full source, inventories, field identifiers and client branding must be preserved. A reduced custom checklist is a deliberate selection, not an accidental truncation of the source.

| Builder step | Expected behavior |
|---|---|
| Choose source | Start from the full master or an accessible saved revision. |
| Select content | Tick individual items or entire sections. Search, select all, clear, mixed section states and live counts support selection. |
| Preview and name | Review included content and provide a clear custom checklist name. |
| Create draft | Create a new unpublished checklist family at version 1. Source IDs, order, groups, options, required metadata and table columns are preserved. |
| Review and publish | Use the existing editor to refine and publish the draft through approval controls. |
| Assign to project | Assign the approved published revision. New surveys freeze its schema and branding. |
| Revise later | Duplicate a locked published revision into an editable version. Existing survey snapshots remain unchanged. |

## Checklist owner acceptance

Before publication, check section coverage, mandatory fields, options, inventory tables, image fields, branding and exported layout. Ensure omissions are intentional. Published revisions are immutable; later template assignments must not rewrite earlier answers or approvals. Project Admin sees only scoped templates and drafts, while Super Admin can govern the full catalog.

## Working checklist behavior

The Engineer enters data against the survey's frozen version. Section review controls support full approval, multi-section partial approval and full rejection. Existing approved sections and Final records remain protected by server checks, not only hidden buttons.

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# Photo evidence workflow

Photo processing should reduce drafting effort while preserving source evidence. Original images remain the reference; AI proposals remain reviewable suggestions until an authorized person explicitly accepts them.

| Stage | Delivered behavior |
|---|---|
| Upload | JPEG, PNG and WebP are decoded and validated. Limits are 20 files per request, 5 MiB per file, 24 megapixels and 200 photos per project/site across its checklists. |
| Preserve | Keep original bytes and filename metadata. Record SHA-256 and detect exact-byte duplicates within a checklist. |
| Prepare | Create a normalized inference preview. Extract only available EXIF GPS and time. Blur and low-detail flags are advisory. |
| Queue | Select photos for a durable background job. Show progress and failures. Retry failed jobs and recover from worker interruption. |
| Analyze | Verify installed local model vision support, then send native base64 image requests to local Ollama. |
| Review | Inspect proposals, reasons, uncertainty, missing information, source photos and suspected anomalies. |
| Accept | Explicit acceptance rechecks authority, field type, original answer and approval locks before changing an answer. |
| Export | Export the JSON evidence report alongside the separate checklist Excel deliverable. |

## Engineer use sequence

Open the assigned site checklist and its photo workspace. Upload relevant, readable evidence in bounded batches. Check the source thumbnails and available metadata, then queue the selected photos. Review each suggestion against the actual photograph and checklist question. Leave unsupported suggestions unaccepted and obtain missing evidence. Submit only after checking the resulting answers.

## Viewer and evidence controls

The workspace includes a large viewer, thumbnails, fullscreen access and source-evidence links. Clients and Final records expose read-only evidence access. Client users cannot upload, enqueue, retry or accept AI drafts. The 200-photo limit is an implemented bound, not a verified throughput or response-time guarantee.

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# Local AI rules and limitations

The application uses local Ollama for operational text and photo inference. Codex development specialists are a separate development mechanism; they are not connected application agents or automatic approval authorities.

## Verified model and local boundary

The installed qwen3.5:4b model reported vision capability through Ollama and passed a native synthetic image probe. The app checks model installation and reported vision support before analyzing photos. Requests use the native loopback API, with no cloud fallback, remote model alias, redirect or proxy path. A missing capability or malformed response fails safely.

## Permitted output

AI can propose values for allowed checklist fields, explain evidence, express uncertainty, request missing information and flag suspected visible anomalies. An anomaly may include a normalized bounding box. Field mapping uses identifiers from the published survey snapshot. Catalog batches cover the source fields without silent truncation.

| Control | Required behavior |
|---|---|
| No fabricated engineering data | Do not invent dimensions, electrical readings, ratings, standards clauses or inspection conclusions. |
| Insufficient evidence | Use unknown or an empty proposal and ask for evidence. Missing EXIF does not become inferred location or time. |
| Protected field classes | Numeric, date, system and photo fields are not automatically filled. Inspection pass and N/A proposals are rejected. |
| Human acceptance | AI does not silently overwrite answers. Acceptance checks stale baselines, permissions, schema and approved sections again. |
| No approval authority | AI cannot submit, approve, reject, close or certify an activity. No detections is not an inspection pass. |
| Controlled inference | Text requests and photo jobs share a database lease. A separate worker provides progress, checkpoints and interruption recovery. |

## Evidence still needed

The completed pilot used synthetic shapes and an explicit acceptance step. Accuracy on user-selected real-site photographs, sustained 200-photo jobs and actual GPU execution or performance have not been demonstrated. A controlled field pilot must measure these before relying on output for project delivery.

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# Reports audit and security

Deliverables should let a reviewer trace an answer to its source and decision history. The Excel checklist and photo JSON report serve different purposes and should be retained together when both are relevant.

| Record | Contents and practical limit |
|---|---|
| Checklist Excel | Site survey, Section approvals and Review history worksheets, with attached checklist-field evidence. Separate photo-pilot originals and anomaly drafts are not automatically embedded. |
| Photo JSON report | Source URLs, image hashes and metadata; model and job details; draft explanations, uncertainty, missing information, anomalies and acceptance provenance. |
| Client decision history | Authenticated identity, timestamps and references are retained in authoritative history and audit. Do not assume generic Excel history columns comprehensively show Client sign-off provenance. |
| Governance audit | Super Admin sees global events; Project Admin sees scoped events. The workspace shows the most recent 500 entries. |
| Original files | Original photo bytes and metadata remain under local media storage. Derived previews do not replace the originals. |

## Implemented access protections

Authentication, CSRF protection, active/lockout checks and project/assignment checks protect actions on the server. Review signatures incorporate the current schema, answers, approvals, routing markers and linked activity state. Invitation paths are redacted from request logs. Approved sections and Final records are protected at mutation endpoints.

## Access maintenance and reset

Super Admin maintenance requires the current password and the typed confirmation RESET ACCESS. It expires sessions, revokes outstanding invitations and clears account lockouts. It preserves users, projects, answers, photos, checklist revisions and audit history. This is an access reset, not a destructive factory reset.

## Security acceptance boundary

Retired global user editors and unsafe legacy deletion tools do not provide an alternate provisioning path. The app currently runs with local development settings. Production hardening, external transport, operational retention policy and independent restore testing remain release work. Never include credentials, invitation tokens or real site evidence in a broadly shared manual.

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# Local operation and data preservation

Keep the checkout and all project runtime data in the approved F drive workspace. The launcher is the normal startup path for the local web app and independent photo worker.

| Component | Location or behavior |
|---|---|
| Project root | F:\iSurvix\iSurvix-Field |
| Startup | Start-iSurvix.ps1 launches the local application and photo worker; the local entry page redirects to the authenticated workspace. |
| Python | Project virtual environment uses Python 3.12.10. Verify this environment when diagnosing startup. |
| Application | Django backend with SQLite, authenticated server-rendered workspaces and checklist/photo views. |
| Runtime data | The project .local folder contains database, media, logs, temporary uploads, secrets, environments and backups. |
| Model service | Ollama runs locally; qwen3.5:4b is the verified vision-capable installed model. |
| Queue execution | Independent photo_worker process uses persistent progress and a shared inference lease, with retry for transient database lock contention. |
| Development specialists | Root orchestration coordinated UX design, backend/security development and API testing; these are development roles, not end-user accounts. |

## Daily operating checks

Open the local page and sign in with the appropriate role. Confirm the intended project and checklist version before assigning or submitting work. Before a photo run, confirm Ollama is running and the selected installed model supports vision. Check queue progress and failure messages. Preserve original files and avoid starting uncontrolled duplicate inference workers.

## Backup and recovery plan

Before a schema upgrade or access change with broad impact, take a consistent database backup and preserve related media, checklist assets and local configuration. Existing checkpoints are retained under .local/backups. Stop application writers for a coordinated restore; do not copy an arbitrary active database over the live one. Validate sign-in, scope, answers, originals and approval history after recovery. A full independent restore drill remains a release acceptance task.

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# Validation and release plan

Delivered means the behavior exists and has supporting checks. It does not mean field accuracy, sustained capacity or production readiness has already been accepted.

| Evidence | Result or remaining limit |
|---|---|
| Automated regression | Earlier full suite: 117 tests passed. Final targeted suite after strict signature change: 43 passed, comprising 17 workflow and 26 builder tests. |
| Workflow controls | Engineer-only submission, Team-Lead-only internal review, Client routing, no-Client closure, stale decisions and historical preservation tested. |
| Builder and responsive UI | Selection, full catalog preservation and desktop/mobile layouts checked; role journeys rendered with synthetic data. |
| Local vision pilot | Native model capability and synthetic upload → queue → AI draft → explicit acceptance → report path passed. |
| Open acceptance | Real-site evidence accuracy, GPU performance, sustained bulk throughput, load latency, email delivery, external access, production security and restore drill remain unverified. |

## Planned delivery gates

| Priority | Next outcome | Acceptance condition |
|---|---|---|
| 1 | Field pilot: Engineer and Team Lead review a controlled real-site photo set. | Trace proposals to originals; measure errors and missing evidence; record human decisions without invented readings. |
| 2 | Operational confidence: run capacity, interruption, backup and restore exercises. | Record observed timing and recovery; preserve evidence and approvals; define supported hardware and batch sizes. |
| 3 | Deployment readiness: choose hosting, authentication, email, transport and support policies. | Approve a separate deployment plan and complete security and user acceptance checks. |
| 4 | Measured sketches: add annotated evidence and editable plan/elevation output. | Require confirmed dimensions and units; show unknowns; validate SVG/DXF output before delivery. |
| 5 | Scale: increase supported workload only after the pilot. | Demonstrate accuracy, bounded resource use and acceptable recovery at the proposed scale. |

**Priorities express a recommended sequence, not a committed schedule. Subproject hierarchy, broader model options and further integrations need explicit product design and acceptance before they are described as delivered.**

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# Reference and acceptance checklist

This guide is a dated application baseline. Source and authenticated operational records take precedence when later changes alter behavior or membership.

## Source map

| Subject | Local source |
|---|---|
| Scope and agreed brief | docs/LOCAL_CODEX_HANDOFF.md; docs/LOCAL_SETUP.md |
| Roles and invitations | docs/ROLES_AND_INVITATIONS.md; backend/core/access.py; governance_views.py; governance_forms.py |
| Stage signatures and review | backend/core/template_views.py; workflow_service.py; models.py |
| Additive routing migration | backend/core/migrations/0019_client_review_routing.py |
| Checklist selection | backend/core/checklist_selection.py; selection_views.py; template_service.py |
| Photo validation and queue | backend/core/photo_service.py; photo_views.py; photo_queue.py; backend/core/management/commands/photo_worker.py |
| Local runtime boundary | backend/checklist/settings_local.py; urls_local.py; Start-iSurvix.ps1 |
| Brand and navigation | backend/static/core/img/logo.png; governance workspace styles and templates |

## Owner acceptance checklist

- Confirm Super Admin can create a project and assign its primary Admin, and an Admin cannot access another project.
- Invite and sign in as Engineer, Team Lead and Client; verify each role sees only its permitted controls and records.
- Build a custom checklist from both sections and individual items; verify source preservation, publication and frozen survey version.
- Run both approval branches: Engineer → Team Lead → Client → closed, and Engineer → Team Lead → closed with no configured Client.
- Exercise partial approval, full rejection, Client rejection and a stale decision; verify locks, open stages and audit evidence.
- Run the real-site photo pilot, inspect originals and proposals, accept only supported answers, then verify separate exports.
- Complete backup and restore validation before broader deployment, and replace or deactivate temporary training accounts.

**Terminology: “Admin” in this guide means Project Admin. “Final” means internally approved; “Closed” means the activity approval chain is complete. “AI draft” means a suggestion, and “demo account” means a requested training identity.**